

Win Probability in IPL (Indian Premiere League)

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Introduction

IPL is a two-month cricket tournament during summer. Ten teams participate in this highly competitive contest, and each team is very competent. It is rare to see a team much weaker than average. We are trying to predict the factors which affects the most for winning a game. Starting out with metrics like sixes, fours, home venue, powerplay runs, and dot balls.

Literature Review

Many studies have researched this topic, each with different aims and methods. After reviewing some published work, two sets of variables stood out as especially important. The first group is about team strength, which includes batting and bowling averages. The second group involves factors related to the toss, such as where the match is played and the time of the game.

Mantri (2021) looked at what matters most for winning IPL matches using 2020 season data. The study used player statistics to measure how strong each team was in batting and bowling. It found that teams with higher batting averages of their top five batsmen were much more likely to win, with a correlation coefficient of +0.8703. In contrast, teams with lower bowling averages for their top two pace bowlers tended to win more, shown by a negative correlation coefficient of -0.786.

For the next set of variables, we looked at studies about ODI (one day international) cricket since we found no studies on IPL which used these metrics. McEwan et al (2023) found that the impact of winning tosses is not generalized across all the international teams. Teams like New Zealand and Australia had the most impact. Whereas Bandulasir (2008) found that toss winning is only effectively contributing when it is a day night game.

These factors are important because of reasons like, the type of bowlers in a team influences their choice to bowl first or second. Pace bowlers dominated teams usually prefer to bowl first, so they can make use of the fresh pitch for speed and swing, before it becomes rough. On the other hand, spin bowling teams like India might want to bowl second, as the worn pitch helps the ball spin more (McEwan et al, 2023).

Whether a match is played during the day or at night matters a lot because dew can form on the pitch and grass during evening games. Dew makes it harder for fielders to catch the ball and can cause the ball to skid more on the pitch, which can change how teams perform in the second innings (McEwan et al, 2023).

These studies were focused on international cricket, which is played worldwide. In contrast, all IPL matches take place in India, where the weather is humid and dew during night games is more likely. IPL schedules two games per day at 3:30 PM and 7:30 PM, which means dew could be a bigger problem for evening matches.

Variable definition

Dependent Variable	Variable Definition
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Bf_won_i	1 if the batting first team of game t wins, 0 otherwise.
Independent Variable (Expected Sign)	Variable Definition
isnight_t (-)	1 if game t is during night, 0 otherwise
bf_sixes_t (+)	Number of sixes hit by batting first team in game t
bf_fours_t (+)	Number of fours hit by batting first team in game t
bf_extras_t (-)	Number of extras conceded by batting first team in game t
bs_sixes_t (-)	Number of sixes hit by batting second team in game t
bs_fours_t (-)	Number of fours hit by batting second team in game t
bs_extras_t (+)	Number of extras hit by batting second team in game t
Bf_pp_margin_t (+)	Number of additional runs scored by batting first team compared to batting second team during Powerplay overs by batting first team compared to batting second team, for game t

Expected signs

Sixes, fours are very intuitive to think. The more boundaries a team scores its better and the less the opposite team scores the better. Extras are vice versa since it is, how many extra runs like penalty runs a team gave away to other team.

Powerplay margin also works the same way, the more you score in the beginning of the game when you had less players fielding outside of infield (more boundaries opportunity) the better it is. We also want to understand how much importance teams are giving during those first six overs since they have one extra bat.

Night games technically benefit, the team who is batting second. Since the temperature and climate in India causes due factor in every other city, the ball is wet and it skids in the pitch. This makes batting easier in the second inning and fielding difficult in the second inning. You can also bowl more effective in the first inning with the dry bowl. We expect (-) coefficient.

Data Section

Our dataset comprises 140 IPL matches played between 2023 and 2025. Data was collected from official IPL records on iplt20.com. We used python to scrape the data. Since 2023, the IPL has added a new impact player rule, where each team can substitute one player for any reason during a match. For example, if a team bats first, they might start with more batsman and one fewer baller. When they ball in the second innings, they can swap a batsman for a bowler. This rule lets batsmen in the top order play more aggressively, as losing a wicket is less risky. Most research about IPL is done before this rule existed, but the impact player rule has made a significant difference to how the games are played now. This is the reason we have only 140 observations available. Previously, 4-5 out of 11 players were bowlers and 6-7 were batsman. Batters had to play slower so they can save their wicket. This rule is making them play more careless and changes the dynamics of the game.

Summary Stats

Table 1					
Variable	Obs	Mean	Std. dev.	Min	Max
isnight	140	.7928571	.406714	0	1
bf_won	140	.6714286	.4713803	0	1
bf_fours	140	15.94286	4.603364	5	28
bf_sixes	140	8.714286	4.363494	1	22
bf_extras	140	9.121429	4.604452	1	25
bs_fours	140	15.2	3.743964	4	25
bs_sixes	140	7.942857	4.182477	0	24
bs_extras	140	8.678571	4.766894	0	22
bf_pp_margin	140	-1.328571	18.63332	-78	54

Summary statistics are presented in Table 1. The dependent variable `bf_won` has a mean of 0.671. This indicates batting-first teams win approximately 67.1% of games in our sample. The batting first team hits an average of 8.71 sixes per match (SD = 4.36), compared to the bowling first team's 7.94 sixes (SD = 4.18). Night games comprise 79.3% of IPL. The powerplay margin variable shows a mean of -1.33 runs, suggesting that on average, batting-first teams lag slightly during the initial overs.

Empirical Model

We estimate three models to examine match outcome predictability. Our baseline specification is a Linear Probability Model (LPM) using OLS:

$$\text{bf_won}_t = \beta_0 + \beta_1 \text{isnight}_t + \beta_2 \text{bf_fours}_t + \beta_3 \text{bf_sixes}_t + \beta_4 \text{bf_extras}_t + \beta_5 \text{bs_fours}_t + \beta_6 \text{bs_sixes}_t + \beta_7 \text{bs_extras}_t + \beta_8 \text{bf_pp_margin}_t + \varepsilon_t$$

The LPM estimates the average change in winning probability (in percentage points) associated with a one-unit increase in each independent variable. Despite LPM having limitations for binary dependent variables (predicted values can fall outside the [0,1] interval), it provides interpretable marginal effects and serves as a useful baseline.

Our second specification is a Logistic Regression Model:

$$\text{logit}(P(\text{bf_won}_t = 1)) = \beta_0 + \beta_1 \text{isnight}_t + \beta_2 \text{bf_fours}_t + \beta_3 \text{bf_sixes}_t + \beta_4 \text{bf_extras}_t + \beta_5 \text{bs_fours}_t + \beta_6 \text{bs_sixes}_t + \beta_7 \text{bs_extras}_t + \beta_8 \text{bf_pp_margin}_t + \varepsilon_t$$

The logit models the log-odds of the batting-first team winning. This ensures predicted probabilities remain bounded in [0,1], making it appropriate for binary outcomes. The coefficients represent changes in log-odds.

Our third specification is the Marginal Effects Model, derived from the logistic regression. Marginal effects measure the change in predicted probability associated with a one-unit increase in each independent variable.

First Model- LPM

Table 2

Source	SS	df	MS	Number of obs	=	140
Model	17.2504864	8	2.1563108	F(8, 131)	=	20.72
Residual	13.6352279	131	.104085709	Prob > F	=	0.0000
				R-squared	=	0.5585
				Adj R-squared	=	0.5316
Total	30.8857143	139	.222199383	Root MSE	=	.32262

bf_won	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
isnight	.0849939	.0703966	1.21	0.229	-.0542673	.2242551
bf_fours	.0266676	.0077293	3.45***	0.001	.0113772	.0419581
bf_sixes	.0584919	.0079996	7.31***	0.000	.0426668	.0743169
bf_extras	-.0300199	.0064135	-4.68***	0.000	-.0427073	-.0173326
bs_fours	-.0384078	.0085606	-4.49***	0.000	-.0553426	-.021473
bs_sixes	-.0460991	.0087127	-5.29***	0.000	-.063335	-.0288633
bs_extras	.0376487	.0062493	6.02***	0.000	.0252861	.0500114
bf_pp_margin	.0018846	.0017267	1.09	0.277	-.0015312	.0053005
_cons	.5687157	.1469083	3.87***	0.000	.2780961	.8593353

There is a one surprising observation, the night variable has a positive coefficient which so far does not make sense. Majority of the variables look statistically significant except bf_pp_margin and isnight.

MultiCollinearity

Table 3

Variable	VIF	1/VIF
bs_sixes	1.77	0.563899
bf_fours	1.69	0.591482
bf_sixes	1.63	0.614573
bf_pp_margin	1.38	0.723372
bs_fours	1.37	0.728969
bs_extras	1.19	0.843799
bf_extras	1.16	0.858689
isnight	1.09	0.913472
Mean VIF	1.41	

we computed variance inflation factors (VIF) for all independent variables in our OLS model. The results appear in Table 3. The VIF for all the variables is way below 5 which indicates no multicollinearity. The mean VIF across all variables is 1.41, well below the threshold, indicating that imperfect multicollinearity is not inflating standard errors or reducing coefficient precision. Thus, we have no evidence of multicollinearity problems.

Heteroskedasticity (Breusch-Pagan)

Table 4

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of bf_won

H0: Constant variance

chi2(1) = 4.80
Prob > chi2 = 0.0285

We used the Breusch-Pagan test on the LPM model. The test statistic was $\chi^2(1) = 4.80$ with $p\text{-value} = 0.0285$, leading us to reject the null hypothesis of homoskedasticity at the 5% significance level. This indicates that error variance varies systematically across observations. Standard OLS standard errors are biased and unreliable.

LPM with Robust standard errors

Linear regression		Table 5		
		Number of obs	=	140
		F(8, 131)	=	33.64
		Prob > F	=	0.0000
		R-squared	=	0.5585
		Root MSE	=	.32262

	bf_won	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
	isnight	.0849939	.0774763	1.10	0.275	-.0682727	.2382605
	bf_fours	.0266676	.0073906	3.61***	0.000	.0120473	.041288
	bf_sixes	.0584919	.0070848	8.26***	0.000	.0444763	.0725074
	bf_extras	-.0300199	.0062777	-4.78***	0.000	-.0424388	-.0176011
	bs_fours	-.0384078	.0078295	-4.91***	0.000	-.0538964	-.0229192
	bs_sixes	-.0460991	.008765	-5.26***	0.000	-.0634384	-.0287598
	bs_extras	.0376487	.0060033	6.27***	0.000	.0257728	.0495247
	bf_pp_margin	.0018846	.0017702	1.06	0.289	-.0016173	.0053866
	_cons	.5687157	.1449993	3.92***	0.000	.2818724	.855559

we estimated the model again using robust standard errors, which remain valid under heteroskedasticity. The results with robust standard errors are presented in Table 5. The coefficient estimates remain unchanged; however, the adjusted standard errors provide reliable inference.

Logit Model with Robust standard errors

Logistic regression		Table 6		
		Number of obs	=	140
		Wald chi2(8)	=	46.11
		Prob > chi2	=	0.0000
		Pseudo R2	=	0.6991

	bf_won	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
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isnight	-.338064	1.108992	-0.30	0.760	-2.511649	1.835521
bf_fours	.459515	.1283545	3.58***	0.000	.2079449	.7110852
bf_sixes	1.018393	.1888545	5.39***	0.000	.6482453	1.388541
bf_extras	-.3593636	.1114135	-3.23***	0.001	-.57773	-.1409973
bs_fours	-.7272895	.1805262	-4.03***	0.000	-1.081114	-.3734646
bs_sixes	-.8122369	.1759363	-4.62***	0.000	-1.157066	-.467408
bs_extras	.621379	.1372846	4.53***	0.000	.3523062	.8904518
bf_pp_margin	.0516314	.0300513	1.72*	0.086	-.007268	.1105309
_cons	2.353593	1.937573	1.21	0.224	-1.443981	6.151167

Logit Model looks more accurate since the isnight variable as appropriate sign.

Marginal Effects (from logit)

Table 7

	Delta-method		z	P> z	[95% conf. interval]	
	dy/dx	std. err.				
isnight	-.0198806	.0541714	-0.37	0.714	-.1260546	.0862935
bf_fours	.0270228	.0061264	4.41***	0.000	.0150153	.0390303
bf_sixes	.0598889	.0076564	7.82***	0.000	.0448826	.0748951
bf_extras	-.0211332	.0048779	-4.33***	0.000	-.0306937	-.0115727
bs_fours	-.0427699	.0092255	-4.64***	0.000	-.0608516	-.0246881
bs_sixes	-.0477654	.0069526	-6.87***	0.000	-.0613923	-.0341385
bs_extras	.0365416	.0063372	5.77***	0.000	.0241208	.0489623
bf_pp_margin	.0030363	.0012741	2.38**	0.017	.0005392	.0055334

Serial Correlation

Table 8

Breusch-Godfrey LM test for autocorrelation

lags (p)	chi2	df	Prob > chi2
4	80.422	4	0.0000

The test statistic is chi-squared(4) = 80.422 with p-value < 0.0001, Therefore we reject the null hypothesis of no serial correlation. This indicates that residuals from consecutive observations are correlated finding consistent. To address serial correlation, we ran the regression using Newey-West standard errors with lag 4. These standard errors are robust to both heteroskedasticity and autocorrelation.

Functional Form (Linktest)

To assess LPM functional form binary dependent variable, we used linktest. This test regresses the original dependent variable on the predicted values ($\hat{y}$) and their squares ($\hat{y}^2$). The significance of $\hat{y}^2$ indicates specification problems.

Table 9

Source	SS	df	MS	Number of obs	=	140
Model	18.6870088	2	9.34350439	F(2, 137)	=	104.93
Residual	12.1987055	137	.089041646	Prob > F	=	0.0000
				R-squared	=	0.6050
				Adj R-squared	=	0.5993
Total	30.8857143	139	.222199383	Root MSE	=	.2984

bf_won	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
_hat	1.903683	.2361793	8.06	0.000	1.436655	2.370711
_hatsq	-.6872026	.1710902	-4.02	0.000	-1.025522	-.3488834
_cons	-.212281	.0758696	-2.80	0.006	-.3623079	-.0622542

The `_hatsq` coefficient is -0.6872 (highly significant, $p < 0.001$), indicating that squared predicted values contribute meaningfully to explaining outcomes. This suggests that the linear model specification is not ideal for our binary dependent variable.

This motivated us to use the logistic regression model, which naturally bounds predictions in $[0,1]$ and is theoretically appropriate for binary outcomes. The logit model's 90.71% correct classification rate confirms it better captures the true data-generating process than the LPM.

Omitted Variables

We considered potential omitted variables that could bias coefficient estimates. Our model's strong explanatory power ($R^2 = 0.5585$, correct classification = 90.71%) suggests we capture the primary determinants of match outcomes.

Irrelevant Variables

To identify variables that do not meaningfully contribute to the model, we examine statistical significance in our preferred specification (logistic regression with robust standard errors, Table 3).

All variables except `isnight` demonstrate statistical significance at conventional levels. The variable `isnight` has a coefficient of -0.3381 with associated z-statistic of -0.30 and p-value = 0.760, indicating no significant relationship between game timing and win probability. This non-significance is surprising given theoretical predictions that evening dew should disadvantage batting-first teams.

We retain `isnight` in the model for several reasons. It represents a theoretically important match characteristic grounded in cricket physics, its inclusion does not substantially affect model fit non-significance and it may represent genuine lack of effect under the Impact Player Rule.

All remaining variables demonstrate statistical significance ($p < 0.05$ or $p < 0.10$ for `bf_pp_margin`), confirming they contribute meaningfully to explaining match outcomes.

Interpretation of LPM Coefficients

In the LPM model,

The coefficient on `bf_sixes` is 0.0585: This indicates that each additional six-run hit by the batting-first team increases the estimated probability of winning by 5.85 percentage points, holding all other variables constant.

The coefficient on `bf_fours` is 0.027: Each additional four-run hit increases winning probability by 2.67 percentage points.

The coefficient on `bf_extras` is -0.0300: Each additional extra run conceded by the batting-first team decreases winning probability by 3.0 percentage points.

The coefficient on `bs_sixes` is -0.0461 : Each six hit by the opposition decreases batting-first team winning probability by 4.61 percentage points, indicating strong opposition offense substantially harms winning prospects.

The coefficient on `bs_fours` is -0.0384 : Opposition fours reduce winning probability by 3.84 percentage points per boundary.

Conclusion

This paper models the match-winning probability for IPL teams batting first, using sample data from 140 matches following introduction of the Impact Player Rule. Three econometric models: linear probability, logistic regression, and marginal effects analysis. All provided consistent evidence regarding key outcome drivers.

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